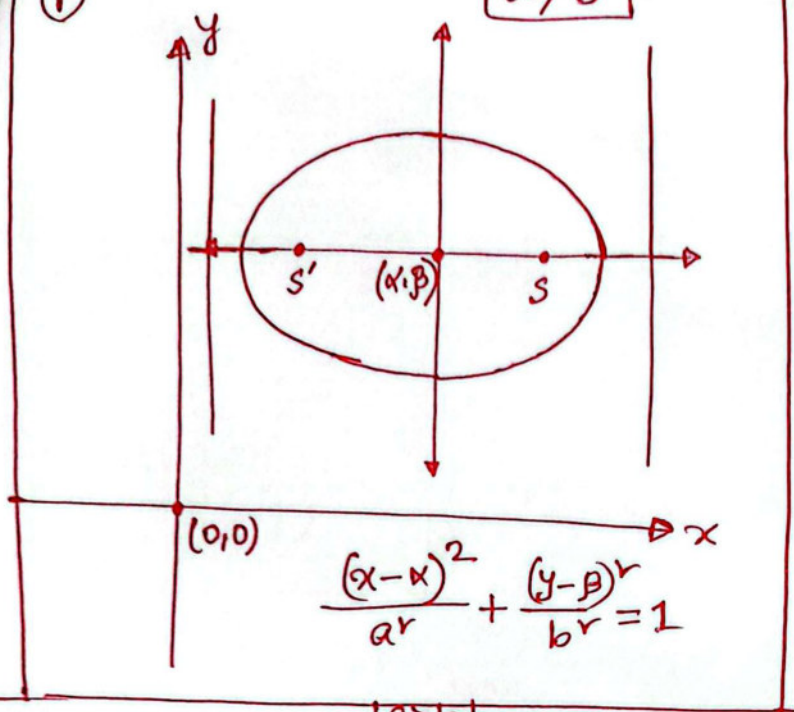
Lec-13

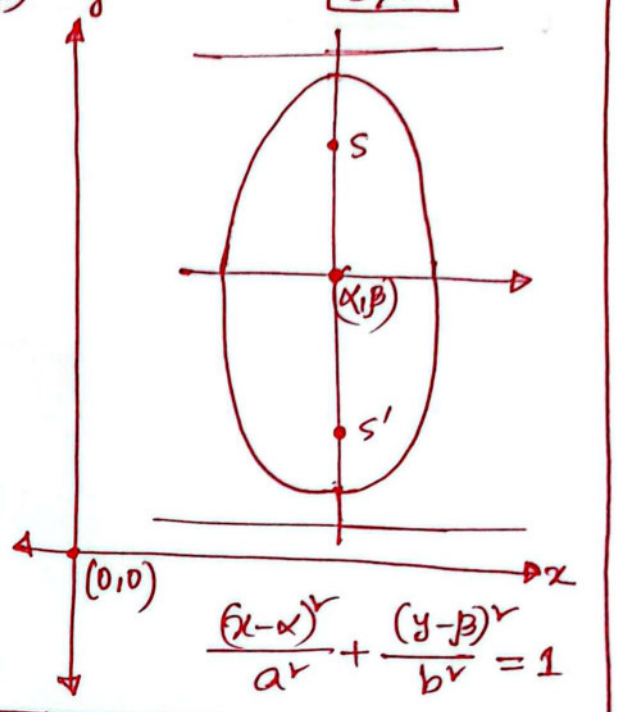
* Necessary formulas for ellipse when centre at (α, β) and (i) $a > b$ (ii) $b > a$

(i) $a > b$



$$\frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$$

(ii) $b > a$



$$\frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$$

	$a > b$	$b > a$
$e = \sqrt{\frac{a^2-b^2}{a^2}}$; $a > b$ $e = \sqrt{\frac{b^2-a^2}{b^2}}$; $b > a$	$\frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$ let, $x-\alpha = X$; $y-\beta = Y$ $\Rightarrow \frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1$	$\frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$ let $X = x-\alpha$; $Y = y-\beta$ then $\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1$
(1) Centre	$X = 0$; $Y = 0$ $\Rightarrow x-\alpha = 0$; $y-\beta = 0$ $\Rightarrow x = \alpha$; $y = \beta$ (α, β)	$X = 0$; $Y = 0$ $x-\alpha = 0$; $y-\beta = 0$ $\therefore x = \alpha$ $y = \beta$; (α, β)
(2) Vertices	$X = \pm a$; $Y = 0$ $\Rightarrow x-\alpha = \pm a$; $y-\beta = 0$ $\Rightarrow x = \alpha \pm a$; $y = \beta$ $(\alpha \pm a, \beta)$	$X = 0$; $Y = \pm b$ $x = \alpha$; $y = \beta \pm b$
(3) Foci/Focuses	$X = \pm ae$; $Y = 0$	$X = 0$; $Y = \pm be$
(4) Eq. of Directrix	$X = \pm \frac{a}{e}$;	$Y = \pm \frac{b}{e}$

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(5) Eq. of Major Axis	$Y=0$	$X=0$
(6) Eq. of Minor Axis	$X=0$	$Y=0$

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(16)

- $e = \sqrt{\frac{a^2-b^2}{a^2}} = \sqrt{\frac{5^2-4^2}{5^2}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

- Focuses; $X = \pm ae$; $Y = 0$

$\Rightarrow x = \pm 5 \cdot \frac{3}{5}$; $y = 0$

$\Rightarrow x = \pm 3$; $y = 0$

$(\pm 3, 0)$

- ~~Vertices~~ centre; $x = 0$; $y = 0$ $\Rightarrow (0, 0)$

- Vertices, $x = \pm a$; $y = 0$

$\Rightarrow x = \pm 5$; $y = 0$

$(\pm 5, 0)$.

- Eq. of latus rectum, $x = \pm ae = \pm 5 \cdot \frac{3}{5}$

$x = \pm 3$

- Eq. of directrix, $x = \pm \frac{a}{e} = \pm \frac{5 \times 5}{3/5}$

$\therefore x = \pm \frac{25}{3}$.

$\Rightarrow 3x = \pm 25$

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7)

- Major Axis; $y=0$
- Minor Axis; $x=0$

(ii) try yourself.

★ <u>Example:</u> Find the vertices and foci, and equations of directrix and sketch its graph of $9x^2-54x+y^2+2y+46=0$

Ans:

$$9x^2-54x+y^2+2y+46=0$$

$$\Rightarrow 9(x^2-6x)+y^2+2y+46=0$$

$$\Rightarrow 9(x^2-2\cdot 3\cdot x+3^2)-9\cdot 3^2+y^2+2\cdot 1\cdot y+1^2 -1^2+46=0$$

$$\Rightarrow 9(x-3)^2+(y+1)^2=81-46+1$$

$$\Rightarrow 9(x-3)^2+(y+1)^2=36$$

$$\Rightarrow \frac{9(x-3)^2}{36}+\frac{(y+1)^2}{36}=1$$

$$\Rightarrow \frac{(x-3)^2}{4}+\frac{(y+1)^2}{36}=1$$

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(18)

$$\therefore \frac{(x-3)^2}{2^2} + \frac{(y+1)^2}{6^2} = 1 \rightarrow (1); \quad a = 2 ; b = 6$$

$b > a$.

Major axis is parallel to Y axis

write (1) putting $X = x-3$; $Y = y+1$

$$\frac{X^2}{2^2} + \frac{Y^2}{6^2} = 1 \longrightarrow (2)$$

- centre: $X = 0$; $Y = 0$ $\Rightarrow x = 3$; $y = -1$; $(3,-1)$
- Vertices ; $X = 0$; $Y = \pm b$

$\Rightarrow x-3 = 0$; $y+1 = \pm 6$

$\Rightarrow x = 3$; $y = -1 \pm 6$

$\therefore y = -7, 5$

$(3,5)$, $(3,-7)$

- eccentricity, $e = \sqrt{\frac{b^2-a^2}{b^2}} = \sqrt{\frac{36-4}{36}}$

$= \sqrt{\frac{32}{36}}$

$= \sqrt{\frac{8}{9}}$

$e = \frac{2\sqrt{2}}{3}$

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- Foci, $X = 0$; $Y = \pm be$

$\Rightarrow x - 3 = 0$; $y + 1 = \pm 6 \cdot \frac{2\sqrt{2}}{3} = \pm 4\sqrt{2}$

$\therefore x = 3$; $y = -1 \pm 4\sqrt{2}$

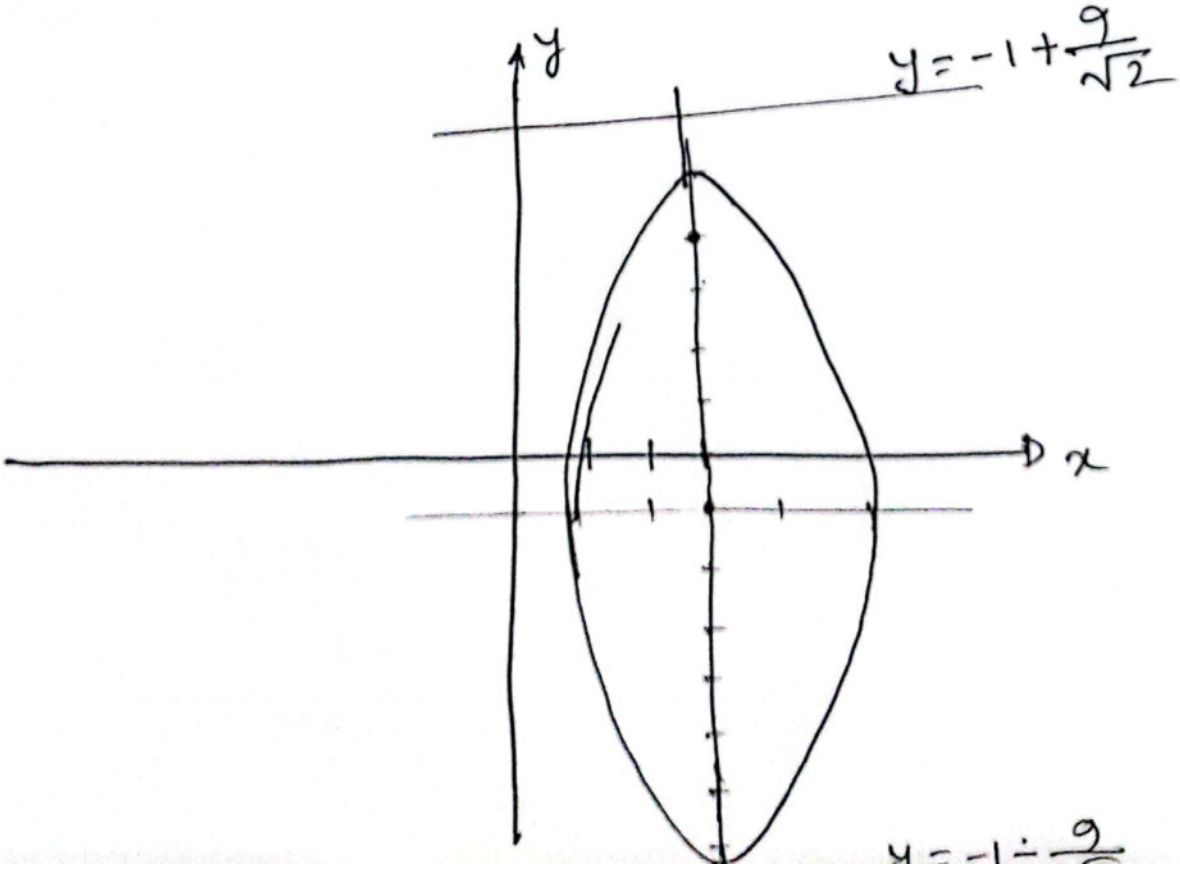
$(3, -1+4\sqrt{2})$ & $(3, -1-4\sqrt{2})$

- Eq. of Directrix,

$$Y = \pm \frac{b}{e} = \pm \frac{6}{\frac{2\sqrt{2}}{3}} = \pm \frac{18}{2\sqrt{2}}$$

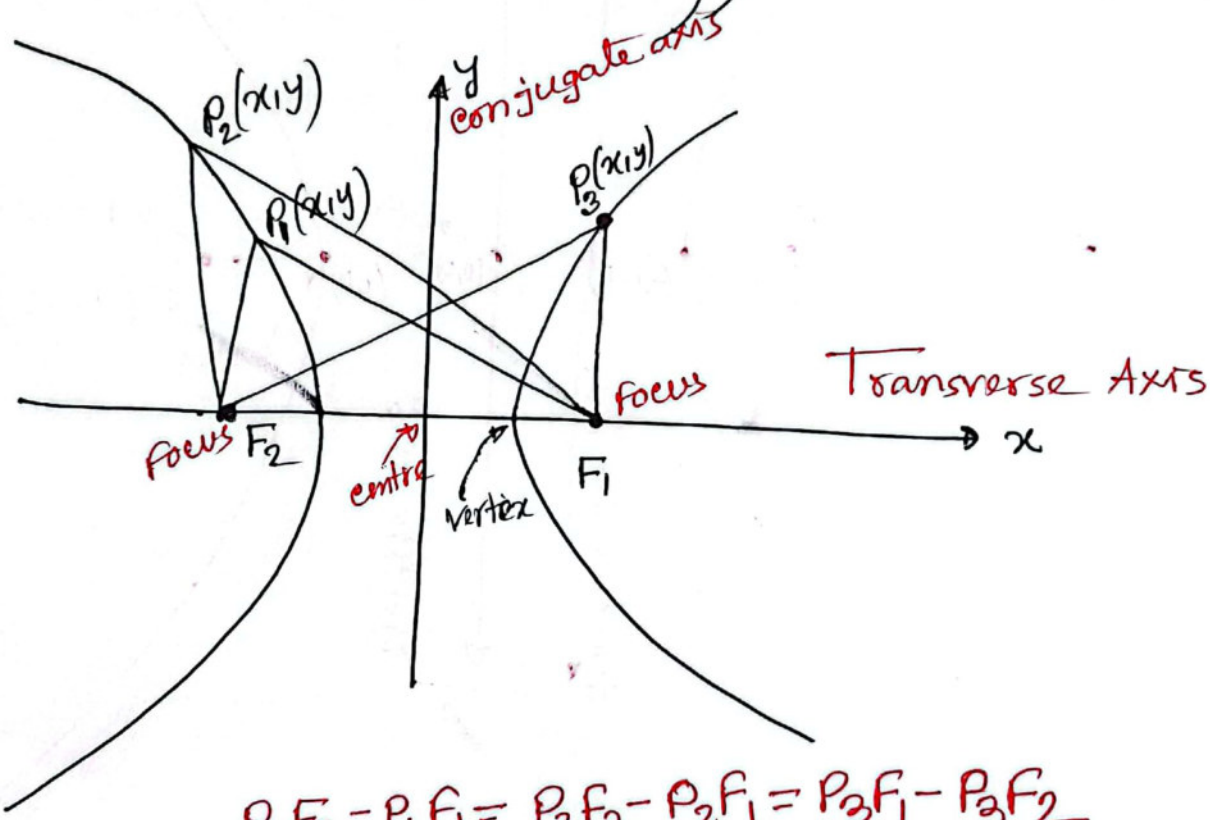
$$\Rightarrow y + 1 = \pm \frac{9}{\sqrt{2}}$$

$$\therefore y = -1 \pm \frac{9}{\sqrt{2}}$$



☆ Hyperbolas:

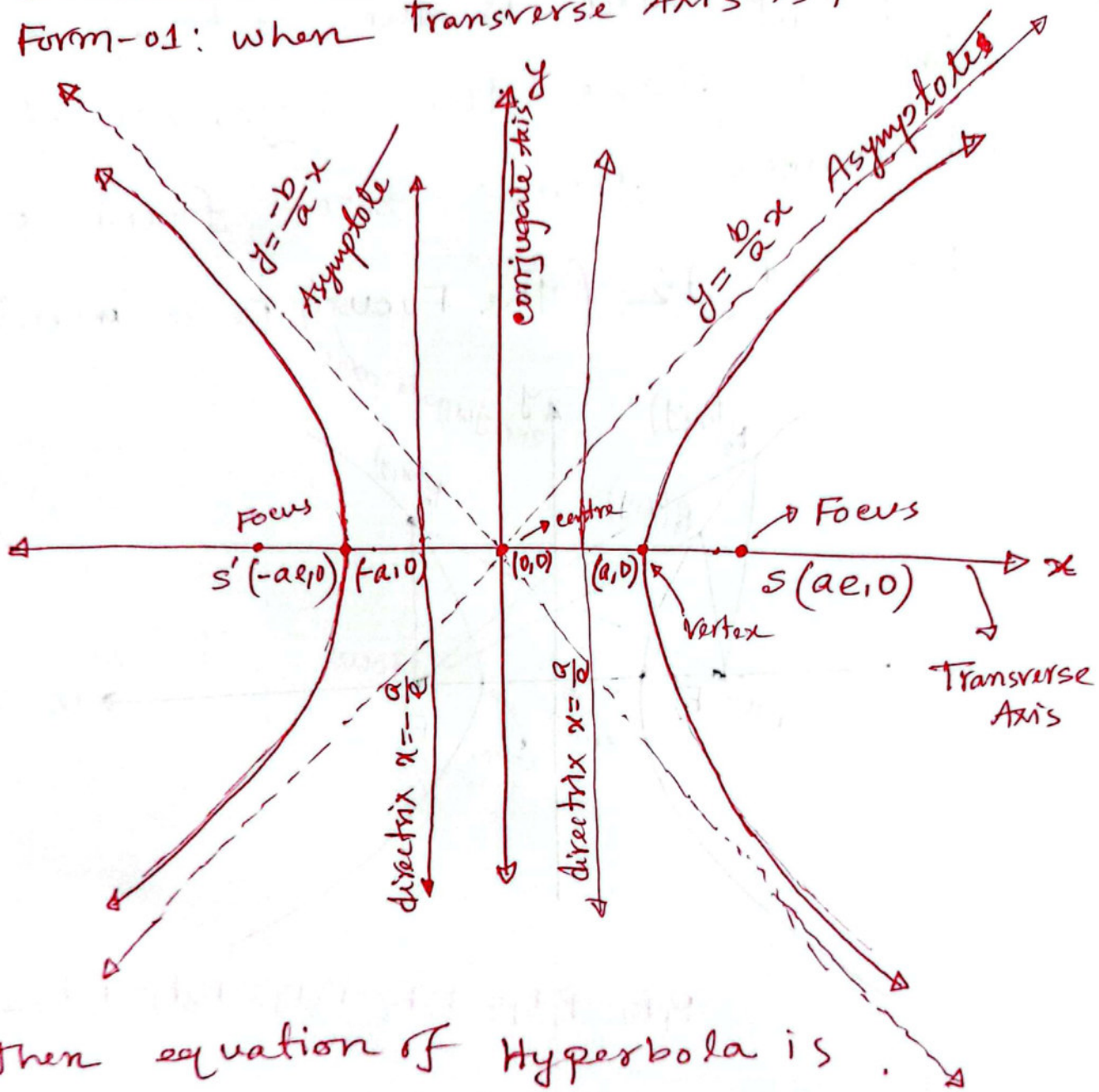
A hyperbola is the set of all points in a plane the difference of whose distances from two fixed points F_1 and F_2 (the Foci(s)) is a constant.



$$P_1F_2 - P_1F_1 = P_2F_2 - P_2F_1 = P_3F_1 - P_3F_2$$

★ standard form of Hyperbola:

Form-01: when Transverse Axis is ~~par~~ X axis.



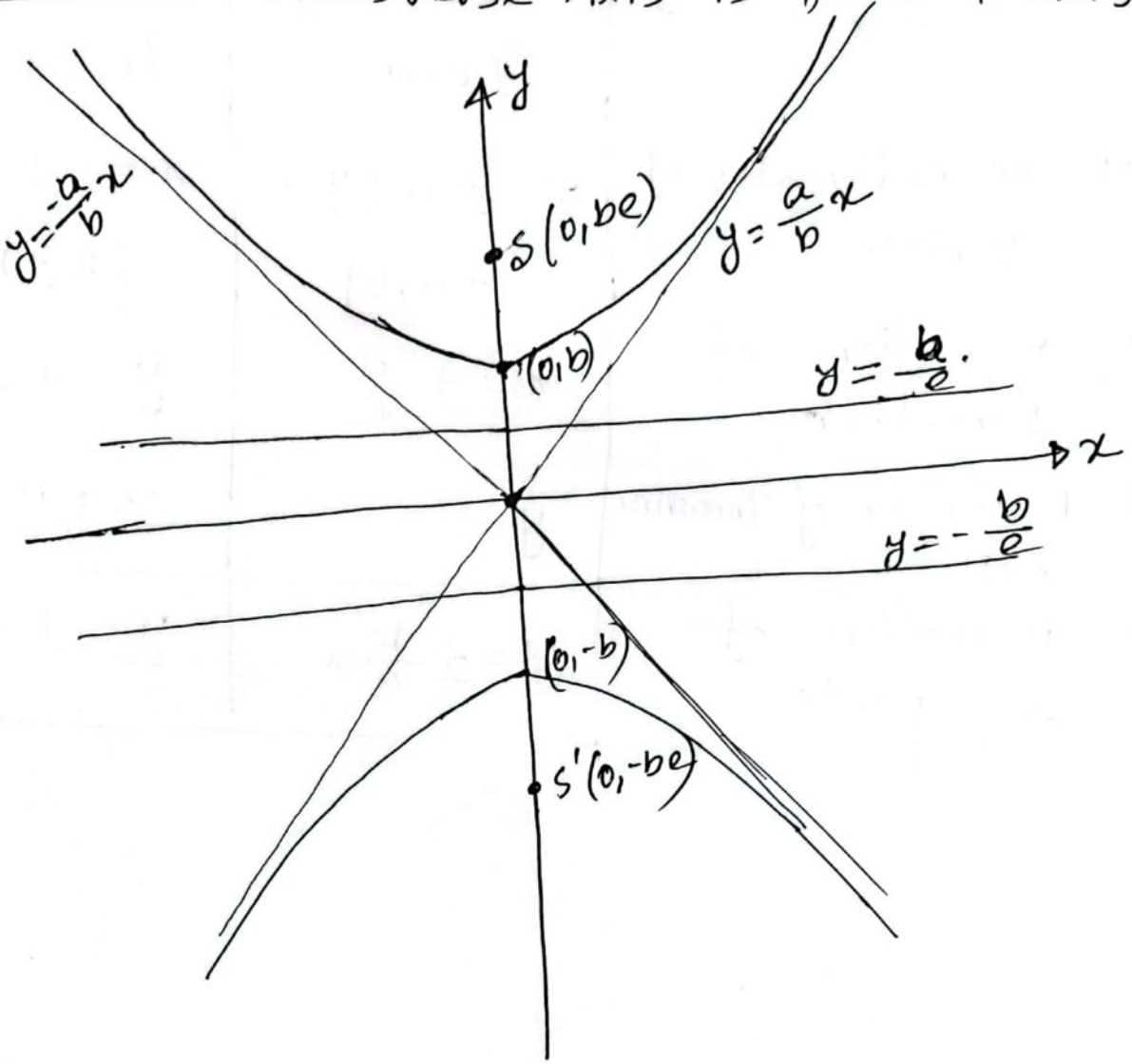
then equation of Hyperbola is.

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 .$$

$\boxed{SP = ePM ; e > 1}$

where, centre $(0,0)$; vertices $(\pm a, 0)$; ~~Focus~~ Foeuses $(\pm ae, 0)$; eccentricity, $e = \sqrt{\frac{a^2+b^2}{a^2}}$

* **Form-02:** Transverse Axis is ~~para~~ Y-axis:



* Then, Equation of Hyperbola : $\frac{y^2}{b^2}-\frac{x^2}{a^2}=1$

* Necessary formulas for Hyperbola:

Description	$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$	$\frac{y^2}{b^2}-\frac{x^2}{a^2}=1$
(1) Coordinates of centre	$x=0$; $y=0$ $(0,0)$	$x=0$; $y=0$ $(0,0)$
(2) Eccentricity	$e=\sqrt{\frac{a^2+b^2}{a^2}}$	$e=\sqrt{\frac{a^2+b^2}{b^2}}$

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3) Co-ordinates of Foeuses	$x = \pm ae; y=0$ $(\pm ae, 0)$	$x=0; y=\pm be$ $(0, \pm be)$
4) Co-ordinates of Vertices	$x = \pm a; y=0$ $(\pm a, 0)$	$x=0; y=\pm b$ $(0, \pm b)$
5) Equations of Directrixes	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e}$
6) Equation of Transverse Axis	$y=0$	$x=0$
7) Equation of Asymptotes	$y = \pm \frac{b}{a} x$	$y = \pm \frac{a}{b} x$

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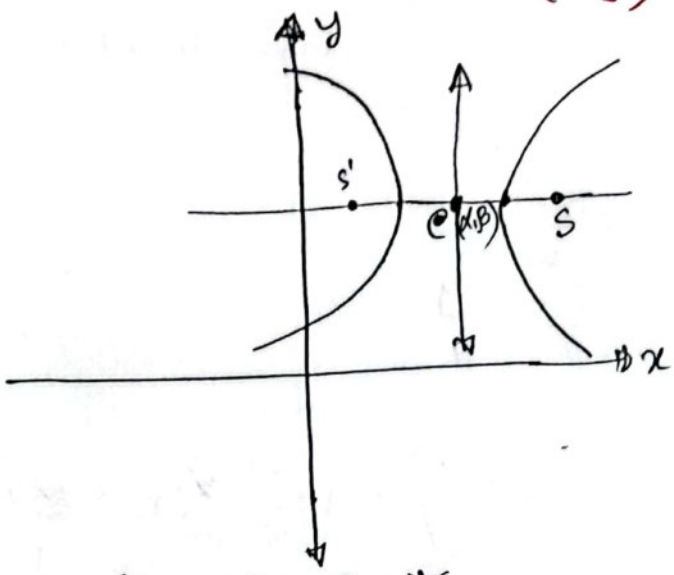
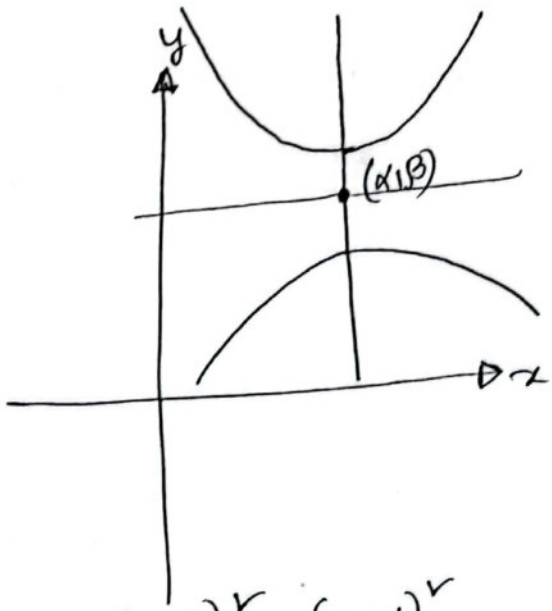
* Necessary formulas for Hyperbola when centre is at (α, β) :





	$\frac{(x-\alpha)^2}{a^2} - \frac{(y-\beta)^2}{b^2} = 1$	$\frac{(y-\beta)^2}{b^2} - \frac{(x-\alpha)^2}{a^2} = 1$
eccentricity, e	$e = \sqrt{\frac{a^2+b^2}{a^2}} > 1$	$e = \sqrt{\frac{a^2+b^2}{b^2}} > 1$
co-ordinates of centre	$x-\alpha = 0 ; y-\beta = 0$ $\therefore x = \alpha ; y = \beta$ (α, β)	$x-\alpha = 0 ; y-\beta = 0$ $\Rightarrow x = \alpha ; y = \beta$ (α, β)
coordinates of Foeuses	$x-\alpha = \pm ae$ $\therefore x = \alpha \pm ae$ & $y-\beta = 0$ $y = \beta$ $(\alpha \pm ae, \beta)$	$x-\alpha = 0 ; x = \alpha$ & $y-\beta = \pm be$ $\Rightarrow y = \beta \pm be$ $(\alpha, \beta \pm be)$
co-ordinates of vertices	$x-\alpha = \pm a$ $\therefore x = \alpha \pm a$ & $y-\beta = 0$ $\therefore y = \beta$ $(\alpha \pm a, \beta)$	$x-\alpha = 0 ; x = \alpha$ & $y-\beta = \pm b$ $\therefore (\alpha, \beta \pm b)$
Equation of Directrixes	$x-\alpha = \pm \frac{a}{e}$ $x = \alpha \pm a/e$	$y-\beta = \pm be$ $\therefore y = \beta \pm be$.

Example

Find eccentricity, co-ordinates of foeuses, equation of the directrices of the hyperbola

$$\frac{y^2}{9} - \frac{x^2}{7} = 1$$

Ans: Given $\frac{y^2}{9} - \frac{x^2}{7} = 1 \Rightarrow \frac{y^2}{3^2} - \frac{x^2}{(\sqrt{7})^2} = 1$

$a = \sqrt{7},\ b = 3$

[Hyperbola of type-2]

$\therefore$ Eccentricity, $e = \sqrt{\frac{a^2+b^2}{b^2}}$

$$\Rightarrow e = \sqrt{\frac{(\sqrt{7})^2+3^2}{3^2}} = \sqrt{\frac{7+9}{9}}$$

$$= \sqrt{\frac{16}{9}} = \sqrt{\frac{130}{81}}$$

$$e = \frac{4}{3}$$

- Foeuses ; $x = 0$; $y = \pm be$

$y = \pm 3 \cdot \frac{4}{3} = \pm 4$

$\therefore (0, \pm 4)$

- Eq. of Directrices, $y = \pm \frac{b}{e} = \pm \frac{3}{4/3} = \pm \frac{9}{4}$

$\therefore y = \pm \frac{9}{4}$.

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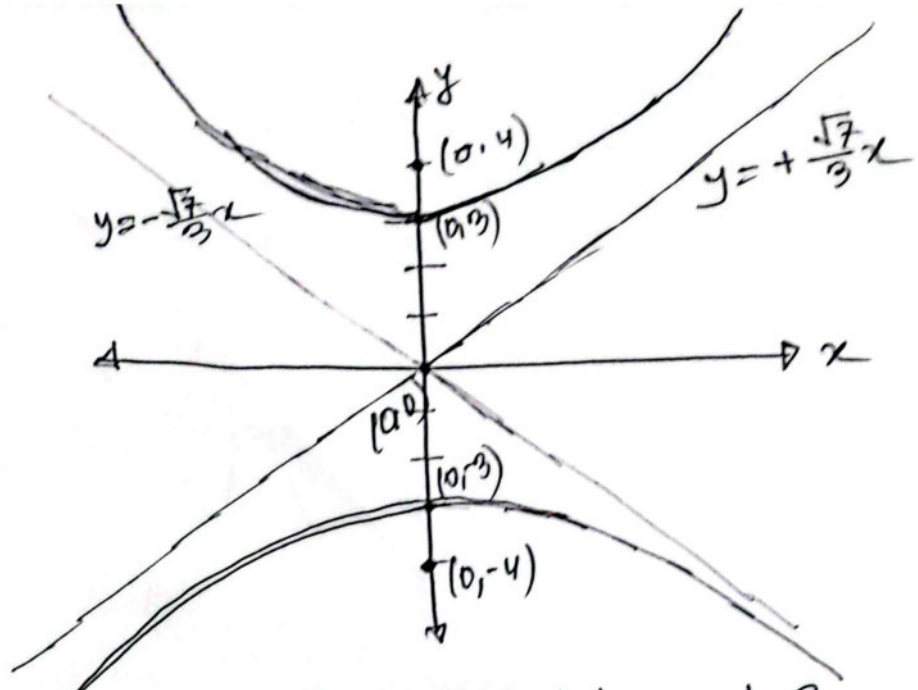


- Vertices, $x=0$; $y=\pm b=\pm 3$ $(0,\pm 3)$
- Asymptotic lines ; $y=\pm\frac{a}{b}x=\pm\frac{\sqrt{7}}{3}x$

★ <u>Example</u>: sketch the hyperbola, and label the vertices, Foceses, eq. of directrices and equation of asymptotes of $16x^2-y^2-32x-6y=57$.

<u>Ans.</u> $16x^2-32x-y^2-6y=57$

$\Rightarrow 16(x^2-2x+1)-16-(y^2+2\cdot 3\cdot y+3^2)+3^2=57$

$\Rightarrow 16(x-1)^2-(y+3)^2=57+16-9$

$\Rightarrow 16(x-1)^2-(y+3)^2=64$

$\Rightarrow \frac{(x-1)^2}{\frac{64}{16}}-\frac{(y+3)^2}{\frac{64}{1}}=1$

$$\Rightarrow \frac{(x-1)^2}{4} - \frac{(y+3)^2}{64} = 1$$ | Hyperbola type (1)

$$\therefore \frac{(x-1)^2}{2^2} - \frac{(y+3)^2}{8^2} = 1$$; $a = 2$; $b = 8$

- Eccentricity, $e = \sqrt{\frac{a^2+b^2}{a^2}} = \sqrt{\frac{4+64}{4}} = \sqrt{\frac{68}{4}}$

$$\Rightarrow e = \sqrt{17}$$

- Vertices, $x - 1 = \pm a = \pm 2$

$\therefore x = 1 \pm 2 \Rightarrow x = -1, 3$

$2y + 3 = 0 \Rightarrow y = -3$

$(-1, -3)$ & $(3, -3)$

- Foeuses; $x - 1 = \pm ae = \pm 2 \cdot \sqrt{17}$

$\therefore x = 1 \pm 2\sqrt{17}$ & $y + 3 = 0$

$y = -3$

$(1 + 2\sqrt{17}, -3)$ & $(1 - 2\sqrt{17}, -3)$

- Eq. of Directrices, $x - 1 = \pm \frac{a}{e} = \pm \frac{2}{\sqrt{17}} + 1$

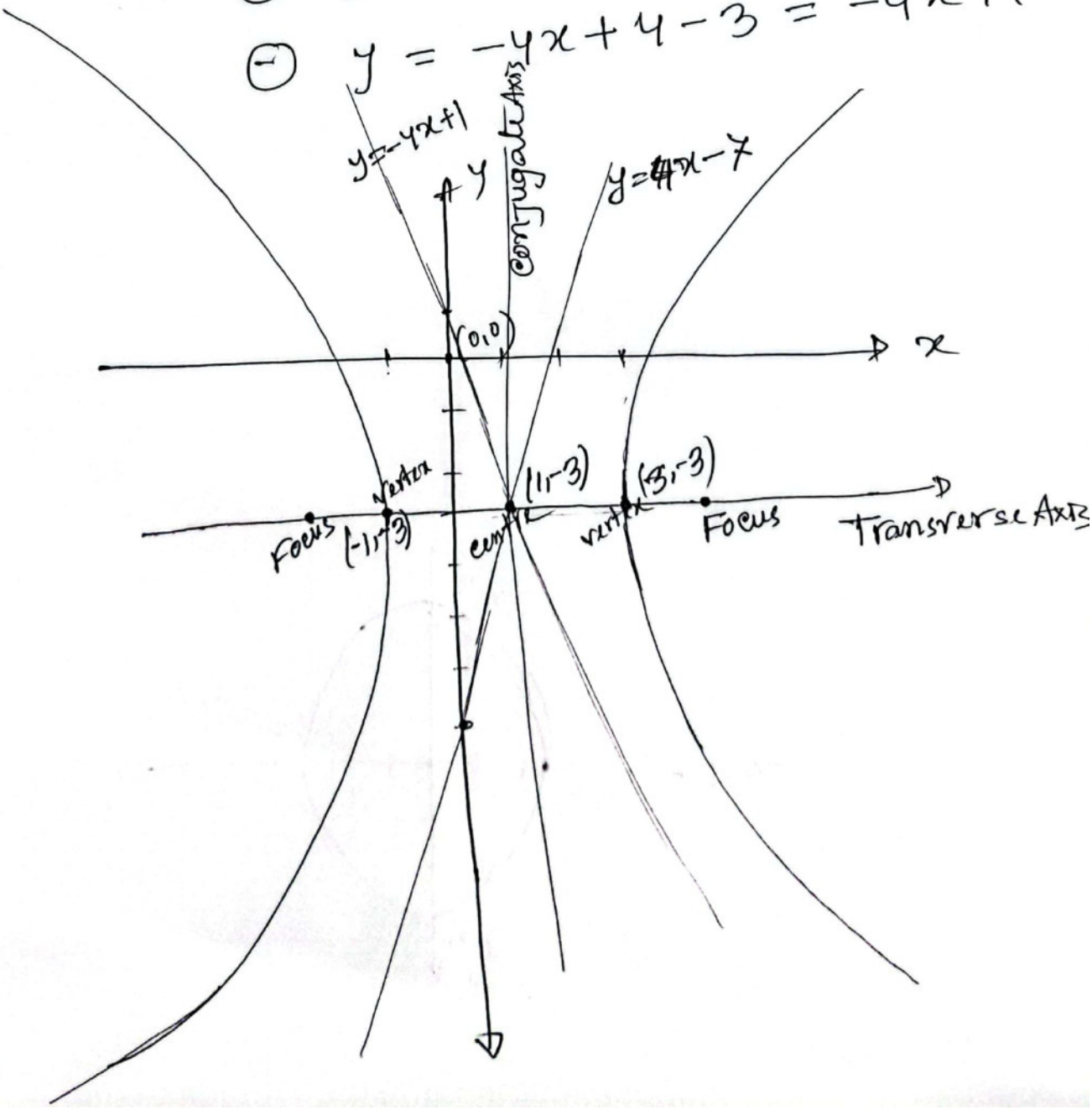
Eq. of Asymptotes

$$y+3 = \pm \frac{b}{a}(x-1)$$

$$\Rightarrow y+3 = \pm \frac{8}{2}(x-1)$$

$$\Rightarrow y+3 = \pm 4(x-1)$$

$$\Rightarrow y = \pm 4(x-1) - 3$$

$$(+) \quad y = 4x-4-3 = 4x-7$$

$$(-) \quad y = -4x+4-3 = -4x+1$$